

Project Documentation

Project Title: Rising Waters

Introduction

Project documentation provides complete information about the development of the Rising Waters project. It includes details of the project objectives, requirements, design, implementation, testing, and deployment. Proper documentation helps users and developers understand the system easily.

Project Description

The Rising Waters project is a Machine Learning-based web application developed to predict flood risk using weather and rainfall parameters. The application accepts user inputs, processes the data using a trained XGBoost model, and displays the prediction result along with the probability of flood occurrence.

Development Process

The project was developed by collecting and preprocessing the dataset, training the Machine Learning model, developing the Flask web application, integrating the prediction model, and testing the complete system before deployment.

Technologies Used

The project uses Python, Flask, HTML, CSS, JavaScript, Pandas, NumPy, Scikit-learn, XGBoost, and Joblib for implementation.

Documentation Contents

The documentation includes project planning, requirement analysis, solution design, implementation details, testing process, source code organization, and project demonstration.

Conclusion

The project documentation provides a complete overview of the Rising Waters Flood Prediction System. It serves as a reference for understanding the project's design, implementation, and overall functionality while supporting future maintenance and improvements.